Fin4120 Fixed Income — One-Page Cheat Sheet

1. Bond Basics / Sectors / Features / Risks

组成: par value, coupon rate, maturity, coupon payment (US通常 semiannual).
US sectors: Treasury(gov, T-bills⩽1yr, mainly IR risk); Agency(GSEs/两房, credit); Municipal(local, tax-free, 1–30+yr, +call risk); Corporate(CP<1yr, +liquidity); ABS/MBS(prepayment); Eurobonds(外币, FX).
China: Gov, Policy Bank, Corporate, Bank, ABS, Panda Bonds. 监管: Interbank vs Exchange; 证监会≈SEC(listed); 银保监; 发改委(SOE bonds).
Structured bonds: designed for specific investor need (derivative / securitization).
Features: Amortizing(分期, safer lump, +reinvest risk) vs Nonamortizing(到期一次) vs Zero(长期目标). Callable→issuer受益; Puttable/Convertible/Exchangeable→investor受益.
Risks — Market: rate, reinvest, inflation, FX, YC, volatility. **Issuer:** credit, call, liquidity, event, tax.
Naming: Tesla 4.5s 1/1/2025 = issuer, 4.5% coupon, matures 1/1/2025. CUSIP = US/CA ID.

2. Pricing / Price–Yield / Floaters

Annuity: FV=C[((1+r)^{n}−1)/r]; PV=C[1−(1+r)$^{-n}$]/r. Bond P = Σ $CF_t/(1+y)^t$.
Yield: rate making PV = market price (moves with price; ≠ coupon).
Price–yield: inverse & convex (同等Δr, P升幅度>P降 → 对investor有利).
Price change drivers: (1) credit (2) market yield (3) **pull-to-par** (P→par as T→0; TVM diminishing).
Floater: coupon = reference + quoted spread. P≈par if required spread≈quoted spread.
Inverse floater: coupon = constant − reference. No-arb: Collateral = Inverse + Floater ⇒ $P_{inv}=P_{coll}-P_{floater}$.

3. Quotes / AI / Dirty–Clean / Day-Count

Treasury quotes in 32nds: 99-16 = 99+16/32=99.5; 99-24+ = 99+24/32+1/64. Corp/Muni often 1/8s.
AI = (days since last coupon / days in coupon period) × coupon. Seller earns daily interest.
Day-count: actual/actual (Treasury); 30/360 (corp, muni).
Dirty (invoice) = Clean (quoted) + AI.

4. Yield Measures / DM / Horizon

EAR = $(1+y_{per})^{m}-1$.
CY = annual coupon / price (ignores ΔP & reinvest). **YTM** includes CG/CL + reinvest assumption.
YTC / YTP / YTS; YTW = min{YTM,YTC,YTP,YTS} (worst for investor). Among them: YTC often highest (call risk), YTP often lowest.
Discount bond: coupon rate < CY < YTM. Premium: reverse.
Portfolio yield: aggregate period CFs, solve one IRR.
Floater DM: (1) assume constant reference (2) try DM (3) discount CFs at ref+DM (4) iterate until PV=price. Limits: static ref; ignores caps/floors. YTM assumes fixed CFs; DM captures floater variability.
Horizon analysis (horizon h): (1) FV of coupons+reinvest @ reinvest rate (2) PV of remaining CFs @ resale yield → sale price (3) Total FV = (1)+(2) (4) solve $P(1+y)^{h}$=Total FV (5) if semi, annualize to EAR. If reinvest rate < YTM ⇒ total return < YTM.

5. Treasury / Auction / WI / STRIPS

Types: T-bills(⩽1yr, discount); T-notes(2–10yr); T-bonds(>10yr); FRN(~2yr); TIPS(5/10/30, fixed real rate).
Primary: competitive (yield+qty) vs noncompetitive (qty only, more aggressive on fill). **Single-price auction:** all pay stop-out (highest accepted yield). Reopening: extra issue of existing (lower cost, more liquidity). Bid-to-cover = public bids / awarded.
Secondary: OTC dealers. On-the-run = most recent; Off-the-run = older.
WI: trade before issuance (price discovery / hedge).
STRIPS: strip coupons+principal into zeros. Uses: duration/CF control; spot curve source; arb-free link coupon↔zeros.
GSE / Agency: housing/农业/基建; GNMA vs Fannie/Freddie. China≈政策性银行债.
Breakeven inflation ≈ $y_{nominal\ Treas} - y_{real\ TIPS}$ ≈ E[inflation]. TIPS nominal ≈ real + actual inflation.

6. Corporate Debt

3 pillars: Bank loans (private); Money-market debt (short); Capital-market debt (bonds/MTN).
Waterfall: Secured ≻ Unsecured ≻ Subordinated ≻ Equity.
Credit vs Default: credit = broader (rating, perception, conditions); default = miss payments. Crisis/rating cut ⇒ credit↑ but default not necessarily.
DLR = d(1−R) where d=default rate, R=recovery. Spread≈compensate DLR (+liquidity/risk premia). IG vs HY: HY higher d & DLR; still can outperform if spread > expected DLR.
Term vs Serial (chunks by maturity). Senior unsecured = promise, no collateral.
Traditional call (par+premium, discount @ market r → issuer-friendly) vs **Make-whole** (PV remaining @ Treas+spread → investor-friendlier).
Sinking fund (retire portion/yr, often random) vs Serial (specified maturities).
PIK: pay interest with more bonds (riskier). **MTN:** "bond vending machine" — high setup, then fast/flexible issuance. Structured note = debt + derivative (customize; mkt/complexity/liq risk).
CP: short-term unsecured, NOT a bond. Bank loans privately negotiated. Syndication: multi-bank one borrower. **CLO:** pool loans → tranches; complexity+leverage = risky.

7. International Bonds

Why: diversify, alpha sources, FX hedge decisions.
Internal (issued&traded in-country): Domestic; Foreign in local ccy (Yankee, Panda, Samurai⋯).
External: Eurobond (outside issuer's home; often bearer/euroccy); Global (multi-market simultaneous).
USD return ≈ local return + FX return; FX↓ hurts unhedged $ investor. Hedge tools: forwards/futures, FX swaps, options; natural hedge (match ccy liabilities).

8. Yield Curve / PCA / Policy

y ≈ market r but can deviate (noise, liquidity, tax).
Shapes: Quasi-flat (stable/uncertain); Increasing (growth/inflation); Decreasing/inverted (recession signal — lock long yields expecting cuts); Humped (mixed / policy shift).
Economic outlook → anticipated CB policy → YC shape.
Theories (brief): Expectations (long = avg expected shorts); Liquidity preference (y(T)=E[r̄]+LPP(T), LPP'>0); Preferred habitat / segmentation (supply–demand by maturity).
Properties: r⩾0; mean reversion dr=a[b−r]dt+⋯; imperfect corr (↓ with maturity gap; short≈1-factor OK); short more volatile than long.
PCA (~95%+): Δy ≈ β_1Level + β_2Slope + β_3Curvature. Level≈parallel; Slope=twist; Curvature=butterfly. Y=$X_{std}V_k$.
PCA steps: standardize → Cov Σ → eigen Σv=λv → sort → V_k → project.
Policy tools: QE (buy long bonds → y_long↓ + liquidity); YCC (pin a tenor); Forward guidance (expectations); IOR (short rates).
YC indicates: health/stress, expectations, risk premia, policy transmission, segmentation (banks short vs pensions long → humps).

9. Zero Curve: Bootstrap / Interp / Spline / NS / Par

DF: for zeros DF(t)=P(t)/F. Spot (annual): z(t)=DF(t)$^{-1/t}$−1. Coupon P=Σ c · DF(t_i)+F · DF(T).
Linear system: P=F · B ⇒ B=F^{-1}P (needs aligned coupon dates).
Bootstrap: use known short zeros/DFs; strip known coupons; solve next DF; roll forward.
Interpolation: linear / cubic spline on DF or spots to fill gaps. **Limit:** linear spot interp assumes z″=0 → miss curvature → coupon-bond mispricing (bootstrap z can differ by tens of bp).
Indirect fit: β=arg min Σ $w_j(P_j-\hat{P}_j)^2$ — smooth & robust, but model misspecification risk.
Spline on DF: piece cubic B(0,t); constraints at knots (value, B′, B″ continuous) + B(0,0)=1. Internalized: B(s)=1+c_0s+b_0s^2+a_0s^3+$(a_1-a_0)(s-1)_+^3$+⋯ (fewer free params). Segments: fit error + complexity penalty.
Nelson–Siegel: R(t,T)=β_0+β_1[(1−$e^{-\lambda\tau}$)/(λτ)]+β_2[(1−$e^{-\lambda\tau}$)/(λτ)−$e^{-\lambda\tau}$], τ=T−t. β_0=level (long), β_1=slope, β_2=curvature, λ=hump location. Fit min Σ$(R_{mkt}-R_{model})^2$; λ nonlinear ⇒ BFGS (or fix λ→OLS). Pros: 4 params, interpretable, smooth. Cons: complex shapes; λ hard; static fit ≠ forecast.
Spline vs NS: spline accurate but can overfit; NS clearer hedge/meaning but may misprice long end.
Par yield: coupon rate that makes price = par (face).